

Time-varying interactions drive species coexistence via rainfall variation

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Abstract: Species interactions are fundamental to maintaining biodiversity. However, how these interactions change over time in response to varying climatic conditions has been poorly explored. This gap limits our ability to identify how temporal changes in coexistence mechanisms sustain diverse communities. Here, we present a modeling approach, coupled with theory, to evaluate how temporal changes in species competitive and mutualistic interactions and intrinsic growth rates promote structural stability. We illustrate our framework using two highly replicated yet independent datasets: (1) annual plant grasslands and (2) wild bee assemblages in shrublands, both subjected to the same variation in rainfall over nine years. We found that both systems respond similarly to year-to-year rainfall changes despite their disparate evolutionary backgrounds. While rainfall variability fosters more diverse communities, prolonged drought leads to substantial biodiversity loss. Although the spatial context dependency modifies some responses, the structural stability of most communities persists through time, with clear winners and losers emerging under environmental variability. The main underlying mechanism determining the observed community stability is a combination of species-specific changes in their growth rates, an increase in the variability of interspecific interactions, and a reduction in the strength of self-limiting effects. Our results underscore the need to consider time-varying species interactions to understand community stability under contrasting climatic conditions and to predict how climate severity influences biodiversity loss.

Significant statement: Environmental changes, such as year-to-year rainfall variation, are expected to influence species persistence. However, it remains unclear how much of these effects arises from shifts in the intrinsic growth of species versus changes in their interactions with other community members. Using two highly replicated studies, we show that interannual rainfall variability alters species' intrinsic growth rates, increases the competition asymmetries between species, and weakens self-limitation. Together, these changes determine the winners and losers in any given rainfall year, suggesting that directional shifts in rainfall could lead to biodiversity loss. Yet, incorporating year-to-year rainfall variation can instead promote diversity by repeatedly reshuffling the temporal identities of winners and losers. Our work introduces a framework for predicting how time-varying species interactions shape biodiversity.

Keywords: temporal networks | time-series data | coexistence metrics | environmental variability | ecological interactions | structural stability

A century of research on biotic interactions has shown that they are not fixed species attributes. Their sign and strength can shift substantially with changes in climate, community composition, and local conditions (Buche et al., 2025; García-Callejas et al., 2021; Ushio et al., 2018), leading to major alterations in the overall structure of species interactions (CaraDonna et al., 2017; Poisot et al., 2015). Although this context dependency is widely recognized (Catford et al., 2022; Chamberlain et al., 2014; Liu and Gaines, 2022; Pearce-Higgins et al., 2015), it has rarely been incorporated into population models that describe the temporal dynamics of ecological communities under natural conditions (Godoy et al.,

2024b; Nguyen et al., 2025). As a result, we still have a limited understanding of how temporal variation in the network of biotic interactions relates to changes in the mechanisms that promote species coexistence.

To address this limitation in a context of climatic variation, it is essential to recognize that species persist in an ever-changing world (Chesson, 2017; Vicente-Serrano et al., 2025), yet such variability is often oversimplified. Logistical constraints are a major reason. Manipulative experiments measuring changes in species interactions, defined as the average per-capita effect of one species on another, are extremely labor-intensive when conducted across more than a few climatic scenarios (Matías et al., 2018; Muehleisen et al., 2025; Van Dyke et al., 2022; Wainwright et al., 2019), making it difficult to capture substantial year-to-year variation. Even when changes in community composition are predicted, they are typically examined at the level of species pairs (Ploughe et al., 2019), focusing on direct interactions while overlooking the indirect positive or negative effects that interaction chains exert on maintaining diverse communities (Levine et al., 2017). Most studies also concentrate on a single taxon or guild, resulting in a limited understanding of how temporally dynamic responses compare across groups with distinct evolutionary histories, such as plants and insects. Additional challenges arise from theoretical simplifications. For example, climate-driven predictions of community composition generally emphasize competitive interactions (Hallett et al., 2019; Van Dyke et al., 2022), neglecting the widespread role of facilitation (Buche et al., 2025; CaraDonna et al., 2021; Losapio et al., 2021) and the fact that shifts from competition to facilitation frequently occur under increasing climatic stress (Maestre et al., 2009).

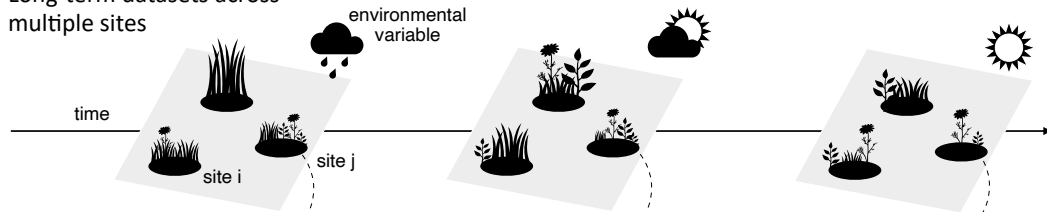
Here, we illustrate how temporal variation in coexistence mechanisms emerges from rainfall-driven changes in species interactions and intrinsic growth rates under natural conditions. We focus on rainfall because prior experimental work has shown that it shapes diversity and community composition through its effects on species' vital rates (Dore, 2005; Knapp et al., 2002; Pearce-Higgins et al., 2015; Sandel et al., 2010; Thibault and Brown, 2008) and on species interactions (Matías et al., 2018; Muehleisen et al., 2025; Van Dyke et al., 2022; Wainwright et al., 2019). To do so, we draw on two highly replicated and multispecies time series: we followed changes in species abundances for (i) 7 annual plants in grasslands during 9 years and across 324 plots and (ii) 4 wild bees in shrublands during 8 years and 40 transects (Fig. S1, and S2, Table S1). In both systems, rainfall θ_t ranged from years with typical rainfall to multiyear droughts, with anomalies of roughly 300 mm below the 85-year average rainfall $\langle\theta\rangle$. For each dataset, we parameterized a Lotka-Volterra-like model in which both intrinsic growth rates $\tilde{r}$ and interaction coefficients $\tilde{A}$ vary through time in response to rainfall (Fig. 1a-c; (2)), while also incorporating temporal variation in spatial stochasticity in $\tilde{r}$ (Supplementary Information S1). To conduct such parameterization, we rely on inverse modeling techniques (Ives et al., 2003). This means that estimated interactions encode neither co-occurrence nor observed interactions (Yin and Rudolf, 2024). They rather quantify the effect of the abundance of species j on the per capita growth of species i (Abrams, 1987; Novak et al., 2016). Although this approach requires rich multi-annual data to estimate interactions with confidence, thereby restricting the number of species to consider, it has the important advantage that we do not impose the sign of interactions. Accounting for both positive and negative interactions sheds light on the ongoing debate over whether environmental variation alters species interactions in predictable ways (Maestre et al., 2009) or whether changes are largely context-dependent (Catford et al., 2022; Chamberlain et al., 2014), with randomness emerging from demographic stochasticity, spatial heterogeneity, unmeasured environmental factors, or simply sheer luck (Shoemaker et al., 2020).

To assess the consequences of temporal changes in coexistence mechanisms for species persistence, we finally embed our modeling approach within the framework of structural stability theory. Structural stability, which can accommodate time-varying model parameters (Godoy et al., 2024b), identifies the parameter ranges that make feasible ecological communities (i.e., all species persist with positive abundances) (Allen-Perkins et al., 2023; Medeiros et al., 2021; Saavedra et al., 2017). Interactions impose constraints, analogous to niche differences (Saavedra et al., 2017), on the space, defined by the vectors of

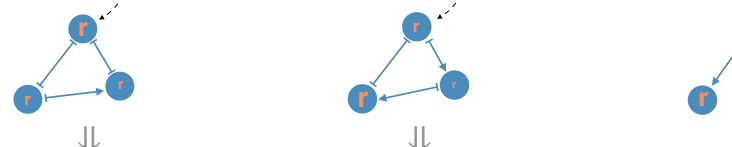
a Time-varying framework

$$\text{abundance change} = \text{time-varying intrinsic growth} + \text{time-varying interactions} + \text{stochasticity}$$

b Long-term datasets across multiple sites



c Time-varying interactions & intrinsic growth rates $\tilde{r}$



d Consequences for stability and diversity

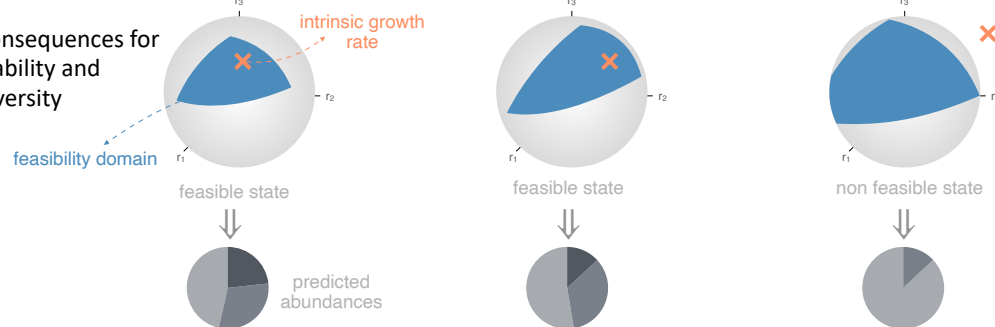


Figure 1: Illustration of the theoretical framework in a toy community of 3 species. (a) Estimation of interactions and intrinsic growth rates that vary through time according to changes in environmental factors. For each dataset, (b) thanks to well-resolved empirical observations from multiple sites constrained to rainfall, we can disentangle the time-varying and stochastic contributions of species' parameters to community dynamics, (c) obtaining time-varying species interactions $\tilde{A}$ (represented as blue networks) and intrinsic growth rates $\tilde{r}$ for each site. (d) In turn, these changes in species parameters modify the emerging properties of ecological communities, such as their stability (represented as the position of $\tilde{r}$, orange cross, within the feasibility domain, blue shape) and diversity.

species’ growth rate, for which all species have a feasible equilibrium. This set is known as the feasibility domain. If the realized growth-rate vector remains within this domain, all species are ensured to persist in the long run (as in the first two time steps of Fig. 1d). Thus, the shape and size of the domain indicate the extent of opportunities for species to coexist (Allen-Perkins et al., 2023; Godoy et al., 2018), and the distance of the intrinsic growth-rate vector to the incenter of the feasibility domain indicates species fitness differences (Godoy et al., 2018). Our main theoretical expectation, when both interactions and species’ intrinsic growth rates vary temporally, is that the degree of structural stability emerges from the interplay between niche differences and fitness differences. In particular, we predict that less structurally stable communities occur in years when rainfall reduces niche differences and increases species’ fitness differences. Ultimately, we hypothesize that, if sustained, the loss of stability will lead to the local extinction of specific species, thereby impoverishing local communities.

Results

Rainfall variability produces contrasting effects on the temporal networks’ structure

Our results show that year-to-year variation in rainfall drives substantial shifts in both the sign and strength of species interactions, affecting interactions among species (interspecific) as well as within species (intraspecific) (coefficients provided in Supplementary Information S2.1). These shifts follow similar patterns in both datasets. To describe the overall time-varying structure of these interactions, we quantified “temporality” (Bauzá Mingueza et al., 2023), (5), a Jaccard-inspired metric that measures how strongly interaction networks differ from one year to another. Temporality is strongly correlated with annual rainfall (Spearman’s $\rho_{plants} = 0.98$; $\rho_{bees} = 0.94$; Fig. S3a) and exhibits comparable trends when calculated using only facilitative or only competitive interactions (Fig. S3b). Rainfall anomalies, defined as deviations from the long-term seasonal average, are positively associated with differences between interaction networks. Thus, large rainfall anomalies lead to markedly different interaction structures, whereas years with minor anomalies show similar interaction networks.

Consistent with the stress gradient hypothesis (Maestre et al., 2009), facilitation is more common in drier years, while competition dominates in wetter years (Fig. 2a). Rainfall anomalies also increase the variance of interspecific interaction strengths. Interactions are weak near the long-term mean $\langle \theta \rangle$ but become stronger and more variable under both drier and wetter anomalies. To evaluate the balance between inter- and intraspecific interactions, we quantified annual “diagonal dominance” ((6)). Theory predicts that diagonally dominant matrices—where intraspecific interactions exceed interspecific ones—stabilize population dynamics and maintain diversity (Chesson, 2000; Saavedra et al., 2017), a pattern widely observed across communities (Adler et al., 2018; Daniel et al., 2024). In both systems, diagonal dominance declines significantly with increasing drought (Fig. 2b), indicating a weakening of intraspecific relative to interspecific interactions. Overall, these findings show that interannual rainfall variation reorganizes the entire structure of species interactions, consistent with observations in other systems (CaraDonna et al., 2017; Domínguez-García et al., 2024).

As intraspecific interactions become relatively weaker with increasing rainfall anomalies in drier years, we would expect a reduction in coexistence opportunities, as measured by the size of the feasibility domain. Yet, we find the opposite. The drier the year, the larger the opportunities to coexist (estimated by the size of the feasibility domain $\Omega(\tilde{A}_t)$, Methods and Fig. 2c). In the driest year (−297mm relative to the average), coexistence opportunities increase for both the annual plant ($R^2 = 0.72$) and wild bee systems ($R^2 = 0.31$). This apparent contradiction is explained by a decrease in intraspecific competition coupled with an increase in facilitation (Fig. 2a). Theory indicates that, in the absence of self-limitation, facilitation between species can alternatively promote coexistence opportunities by preventing the ex-

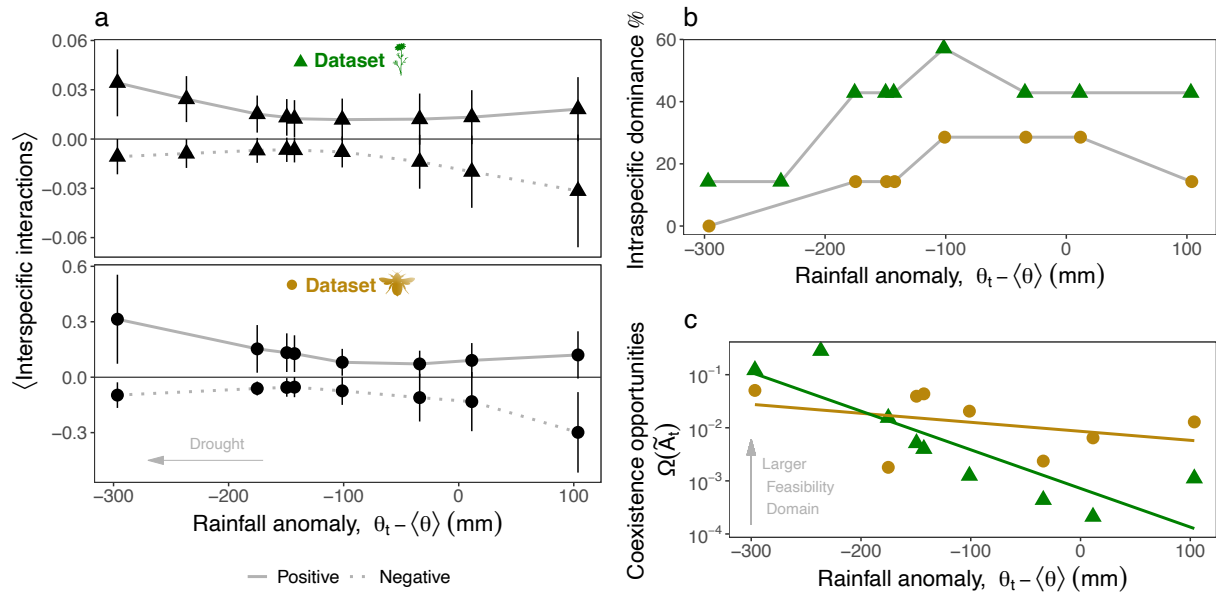


Figure 2: **Structural changes in interaction networks over time.** **a** Mean interspecific positive (solid) and negative (dotted) interactions over species and sites versus rainfall anomalies. **b** Changes in the percentage of species whose self-regulation dominates their interspecific interactions with rainfall anomalies. Smaller values indicate less self-regulation. Gray lines for visualization guidance. **c** Coexistence opportunities (size of the feasibility domain for each realized interaction matrix $\Omega(\tilde{A}_t)$) with respect to rainfall anomalies for our two independent datasets. Solid lines are regressions (Table S2). Supplementary information S2 contains the interaction histograms, values, and standard deviations of the time-varying networks.

tion of poor-performing species that persist thanks to the presence of others (Godoy et al., 2024a; Rohr et al., 2014). Therefore, our results suggest that, in years of low rainfall, our two case studies could potentially withstand a broader range of intrinsic growth-rate variation thanks to the prevalence of positive interspecific interactions (Gross et al., 2024). However, as noted above, it is essential to compare the magnitude of coexistence opportunities (i.e., niche differences) with the differences in species' intrinsic growth rates (i.e., fitness differences) under each rainfall scenario for this statement to hold (Fig. 1d).

Rainfall variability shapes species performance, determining winners and losers

Interannual changes in rainfall affected the environment-driven intrinsic growth rate $\tilde{r}(t)$ in predictable ways, though the direction of these effects differed between datasets. In the plant community, intrinsic growth rates generally increase with positive rainfall anomalies (wetter years), whereas wild bees showed the opposite trend (Fig. 3a). This contrast reflects fundamental ecological differences. Plants depend on water availability for germination, growth, and reproduction (Knapp et al., 2002; Wainwright et al., 2018), while ground-nesting wild bees may experience reproductive disruptions and impaired offspring development under soil-flooded conditions (Harmon-Threatt, 2020). However, we also observed exceptions in both systems—for example, *Beta macrocarpa* and *Flavipanurgus venustus*. Such asymmetric responses are consistent with previous studies showing that species do not benefit equally from environmental conditions, leading to winners and losers under changing rainfall regimes (Knapp et al., 2002; Matías et al., 2018; Thibault and Brown, 2008; Van Dyke et al., 2022; Wainwright et al., 2019). Intrinsic growth rates do not depend solely on rainfall. Site-level stochasticity (spatial heterogeneity) modulates the

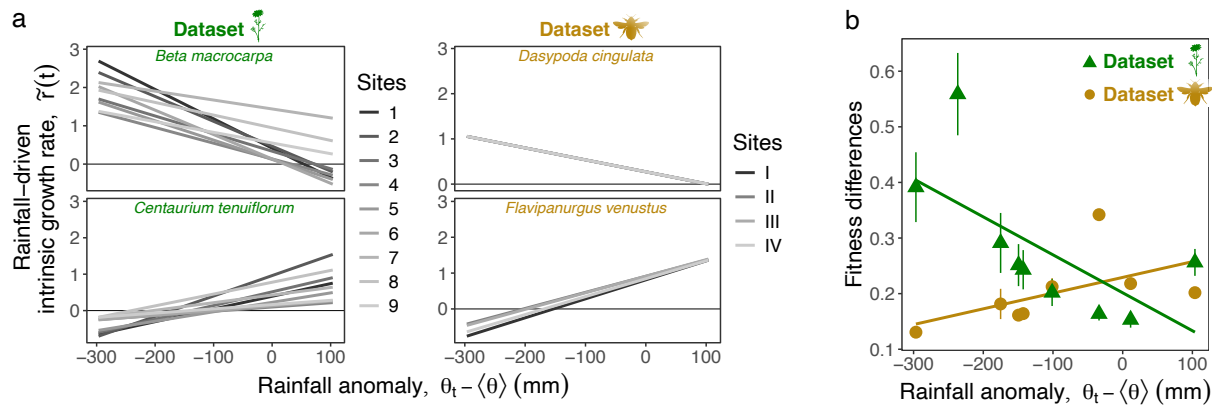


Figure 3: Species' performance explained by deterministic effects of rainfall and spatial context dependency. **a** Realized intrinsic growth rates for all sampled sites. A relationship parallel to the x-axis indicates no sensitivity to rainfall anomalies, and a greater slope variation for a species means more sensitivity to the site's conditions. *D. cingulata* was found in every site, but its realized intrinsic growth rate $\tilde{r}$ was not sensitive to site. Fig. S5 and S6 contain the remaining species. **b** Fitness differences associated with rainfall anomalies, measured as the distance between $\tilde{r}$ and the incenter (Methods). Each point is averaged across sites. Goodness of fits are in Table S2 and estimated terms in Tables S5 and S6.

realized intrinsic growth rates $\tilde{r}(t)$, either amplifying or dampening rainfall effects (gray lines in Fig. 3a). Consequently, some sites buffer poor years while others intensify negative impacts, with this site-to-site variability being greater in the plant dataset than in the wild bee dataset (Fig. S4). The cause of this taxonomic difference remains unresolved. One possibility is sampling bias, as plant populations are sampled more intensively than bee populations, and the larger mobility of bees could homogenize responses across sites. Future work should test whether certain taxa are inherently more sensitive to stochasticity in their responses to rainfall variation.

What our study reveals, however, is that these differences in how rainfall drives species' intrinsic growth rates promote fitness differences. We quantified these differences by measuring the distance between the intrinsic growth rates of all species and the vector for which extinctions were less probable, the incenter of the feasibility domain (Saavedra et al., 2017). We find that drier years increase fitness differences for the plant community ($\rho = -0.67$), yet it does the opposite for the wild bee community ($\rho = 0.54$) (Fig. 3b). This is consistent with plants performing better and more similarly in wetter years and with wild bees performing better in drier years (Fig. S5 and S6). A direct prediction from this finding is that the annual plant community is more prone to species loss during dry years, while the wild bee community is more vulnerable during wet years. The underlying mechanism explaining this prediction is that the more similar the performance across species, the greater the likelihood of persistence because the community is more centrally located within the feasibility domain, and farther from its exclusion borders (Allen-Perkins et al., 2023; Nguyen et al., 2025; Saavedra et al., 2017).

Assessing structure stability under time-varying interactions and intrinsic growth rates

Building on our earlier results, we examined how structural stability emerges from the interplay between time-varying coexistence opportunities (Fig. 2c) and differences in species' intrinsic growth rates (Fig. 3b). That is, the balance between niche and fitness differences. We quantified structural stability as the distance between a species' intrinsic growth rate and the closest exclusion boundary within the feasibility domain (Fig. S7) (Allen-Perkins et al., 2023; Lepori et al., 2024; Medeiros et al., 2021). Al-

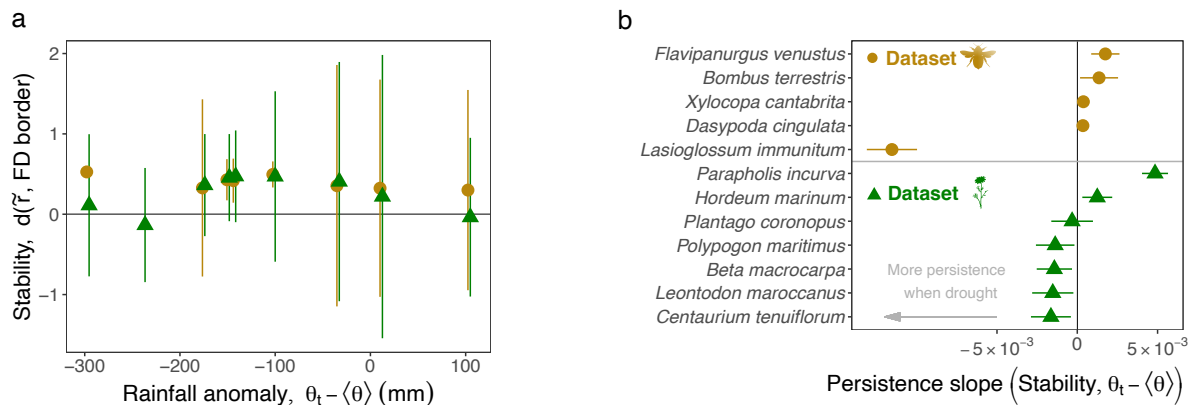


Figure 4: **Consequences of drought on stability.** **a** Structural stability as the distance between intrinsic growth rates and the nearest border of the feasibility domain (FD) where a species goes extinct. Points are averaged across species and sites and are jittered to avoid overlapping. **b** Slopes of persistence for each species as a function of rainfall anomalies. The trends for each species are in Fig. S8.

though coexistence opportunities increase under lower rainfall, asymmetries in intrinsic growth rates $\tilde{r}$ also become more variable, producing no clear directional trend at the community level (Fig. 4a). Thus, rainfall deviations neither strengthen nor weaken structural stability overall. Instead, the spatial heterogeneity documented above becomes a major source of variability. The wide error bars in Fig. 4a show that some sites maintain positive structural stability and help preserve diversity, whereas others sit near the stability boundary and risk species loss. Some sites even show negative structural stability, indicating expected extinctions due to declining populations.

To identify which species are more prone to extinction under greater fitness differences, we computed the slope between rainfall anomalies and the distance to the edge of the feasibility domain in which each species faces extinction (Allen-Perkins et al., 2023; Lepori et al., 2024; Medeiros et al., 2021). For our study, positive distances denote that a species is more prone to persist during wetter years, and negative slopes indicate greater species persistence during drier years (Fig. 4b). Because most species exhibit a null or very weak slope (i.e., values centered near zero), this suggests that the observed fitness differences at the community level are driven by a few specific species whose persistence, influenced by rainfall preferences, deviates from the rest. *Parapholis incurva* is a plant example, whose persistence is closely linked to wet years, resulting in poor performance during dry years.

A surprising result is the weak relationship between changes in rainfall and species persistence (Fig. 4b). One might expect species to persist best under rainfall conditions that maximize their performance, either by avoiding competitive exclusion or environmental filtering, but this pattern does not emerge. Although species show clear preferences for dry or wet conditions in the absence of interactions (Fig. 3a), their persistence in the full community is highest under conditions close to the long-term average, not at extremes. The most paradigmatic example is the annual plant species *Centaureum tenuiflorum*. Despite obtaining higher intrinsic growth rates in wetter years, its predicted persistence increases slightly under low rainfall. This discrepancy between direct and realized responses indicates that interaction networks strongly reshape species' climatic niches. While the role of biotic interactions in modifying climatic niches has been discussed at biogeographic scales (Galiana et al., 2023; Stephan et al., 2021; Wiens, 2011), it has rarely been evaluated locally. Our results show that the complexity of interaction networks prevents species from capitalizing on extreme rainfall conditions; instead, most species persist best under rainfall near the long-term mean.

Predictions of biodiversity loss and the amplification of spatial context dependence

Theory predicts that falling outside the region of coexistence opportunities, reflected in negative structural stability values (Fig. 4a), can have major consequences for local biodiversity (Allen-Perkins et al., 2023; Godoy et al., 2018; Lepori et al., 2024; Saavedra et al., 2017). Our time-varying approach reveals the mechanisms underlying this potential loss. Rainfall changes the performance of particular species (Fig. 3a), increases the variability of both positive and negative interspecific interactions (Fig. 2a), and weakens self-limiting effects (Fig. 2b). Crucially, this framework also enables us to predict how biodiversity will change under different rainfall regimes.

We start by exploring a case with constant rainfall, equal to the historical average $\langle \theta \rangle$. Although real environmental conditions fluctuate continuously, this baseline provides a useful reference for comparing other rainfall regimes (see Methods). We then explored three additional regimes: a gradual decrease in rainfall ($\downarrow \theta_t$), a gradual increase ($\uparrow \theta_t$), and an oscillation between minimum and maximum values ($\downarrow \uparrow \theta_t$). For each regime, we simulated population dynamics using the parameterized model for the corresponding time-varying rainfall levels and calculated the Shannon diversity index annually (Fig. 5a) and at the end of the simulations (Fig. 5b). These simulations have three possible outcomes. (i) the presence of a persistent group of species (positive Shannon diversity), (ii) the complete species extinction of the community, or (iii) the dominance by a single species (in the last two cases, Shannon diversity is zero).

For most sites in both study systems, we find that the constant, increasing, and oscillating rainfall regimes have a positive Shannon diversity at the end of the simulations (Fig. S9), consistent with the empirical datasets. In the wild bee dataset, only the $\uparrow \theta_t$ regime prevents the community from collapsing. Note that the documented rainfall does not change gradually but instead shows peaks and troughs—despite the recent drought trend (Fig. S2), a pattern resembling the oscillating regime. This oscillating scenario also produces the highest Shannon diversity in the plant system. By contrast, the gradual drought regime ($\downarrow \theta_t$) typically drives diversity to zero. This highlights that although context-dependency (i.e., differences among sites) influences biodiversity loss, the mechanisms we identified can maintain coexistence only under certain environmental fluctuations. In contrast, persistent drought leads to a severe decline in biodiversity.

Conclusions

In summary, our results provide deep insights into how ecological communities respond to time-varying environments, particularly, temporal changes in rainfall. By adopting temporal networks of interactions alongside time-varying intrinsic growth rates, we show that a potential biodiversity loss arises from combining relative increases in the performance of particular species, an increase in the variability of interspecific positive and negative interactions, and a reduction of the strength of self-limiting effects. Accounting for this temporal variability also reveals unanticipated shifts in niche structure and fitness differences that are not apparent from static snapshots. Beyond retrospective explanation, our tractable approach enables the simulation of other scenarios, in contrast to methods based on machine learning or instantaneous attractor reconstructions, e.g., S-maps; (Liu and Gaines, 2022; Nguyen et al., 2025; Ushio et al., 2018), that often preclude mechanistic interpretation and require fine-tuning.

A few caveats are worth noting. Our study systems comprise communities with annual life cycles, which facilitate inference with the available time series. For communities with generation times spanning multiple years, denser or longer sampling will be necessary. More generally, the mechanistic structure of our model requires substantial data, especially in larger communities, which can limit applicability and may require reframing the scope and the questions posed about temporal variability in those systems.

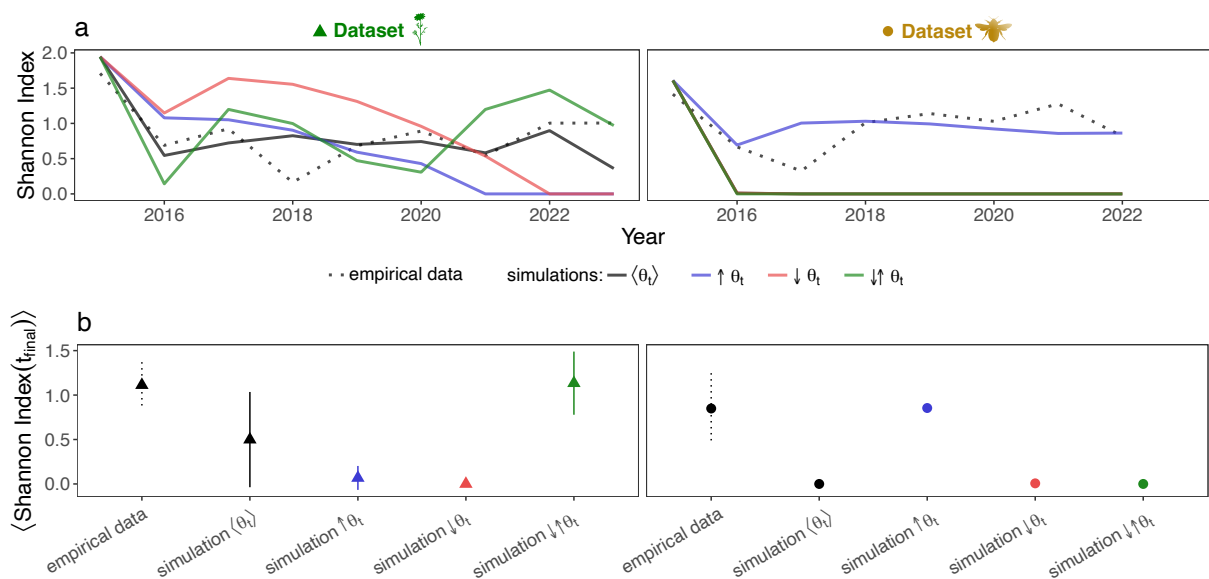


Figure 5: **Consequences of drought on diversity.** **a** Shannon diversity index, (7), for different rainfall regimes. Solid lines show simulations from our estimated model for different scenarios: rainfall does not vary (black line), increases over time (red), decreases (blue), and oscillates between the maximum and minimum values over 9 years (green). The sites simulated are site 2 for the plant community (left) and site I for the wild bees (right). Simulations in the latter dataset overlap, except for the increase in rainfall. Fig. S9 shows the remaining sites. **b** Shannon indices for the final state of the simulations. Error bars represent the standard deviations across all sites.

Despite the demands, the model reproduces growth trajectories (Fig. S10), supporting the adequacy of the chosen formulation and indicating that it captures the effect of environmental variability on species dynamics.

To our knowledge, none of these results has been reported previously in experimental or observational studies, largely due to theoretical and experimental limitations, making our study a benchmark for understanding the effects of reduced rainfall under natural conditions. In addition, although our two systems differ in soil heterogeneity, spatial extent, and other features that can modulate the magnitude of context-dependent changes in growth and interactions, and thus in species persistence, we identify common mechanisms across these communities of contrasting evolutionary histories. These commonalities open the door to designing conservation strategies that prevent biodiversity loss amid increased rainfall anomalies.

Methods

Time-varying framework

Our theoretical approach is based on the Ricker model, which is a discrete version of the generalized Lotka-Volterra equations (Levine and HilleRisLambers, 2009), where changes in species i abundances at time t (N_t^i) are classically described by:

$$\log \frac{N_{t+1}^i}{N_t^i} = r_i + \sum_{j=1}^n A_{ij} N_t^j \quad (1)$$

with parameters accounting for intrinsic growth rate of species i r_i and interaction matrix A (Novak et al., 2016) of the n species. This deterministic model does not depend on any temporal variable. The Ricker model is commonly applied to plant communities (Mayfield and Stouffer, 2017); for completeness, we also apply it to the wild bee community. Notably, it accommodates both positive and negative interactions, unlike the standard Beverton–Holt formulation. We introduced time-varying species responses in both intrinsic growth rates r'_i and interactions B_{ij} via effects of environmental conditions at each time. For our two assemblages, the driver on which they depend is rainfall θ_t . Moreover, we modeled the local contingencies of the different sampling sites as a stochastic effect on species' performance (species' intrinsic growth rates). This effect modifies the per capita growth of species by $u_{i,b}$. Our proposed framework reads:

$$\log \frac{N_{t+1}^{i,b}}{N_t^{i,b}} = r_i + r'_i \theta_t + \sum_{j=1}^n (A_{ij} + B_{ij} \theta_t) N_t^{j,b} + u_{i,b} \quad (2)$$

where the subscripts b represent different sampling sites. The time-varying terms of (2) involve a deterministic term on the growth rates $r'_i \theta_t$ and interactions $B_{ij} \theta_t$, and a stochastic growth term $u'_{i,b} \theta_t$. Ultimately, the realized interactions and intrinsic growth are a combination of all those factors:

$$\tilde{A}_{t,ij} = A_{ij} + B_{ij} \theta_t \quad (3)$$

$$\tilde{r}_{t,b,i} = r_i + u_{i,b} + r'_i \theta_t \quad (4)$$

Hence, species interactions change annually due to our selected environmental variable, and species exhibit different performances depending on annual rainfall and the site b they have been sampled.

We estimated the parameters using the R nlme package for generalized linear mixed-effects models, which allows the separation of deterministic and stochastic effects. The statistical power stems from the fact that sampling was conducted across several sites, which were divided into subplots. We also

fitted other time-dependent models (see discussion in Supplementary Information S1 and Table S3), but (2) provided the best trade-off between being complex enough and offering statistically better fit, and validation of predicted abundances (Fig. S10). To assess the relative quality of our models, we calculated the Akaike Information Criterion (AIC). It balances the goodness-of-fit with the number of estimated parameters to prevent overfitting. We compared the AIC of the simplest model, (1), with that of models including stochastic and time-varying effects (Fig. S11 and Table S4). Goodness-of-fit and correlation coefficients were calculated with the R Stats package. Even though the multi-annual plant dataset observational studies would allow the inclusion of a stochastic term in the interactions, doing so did not qualitatively change our results. It just created variability over the same pattern of interactions.

To disentangle the contribution of deterministic and stochastic effects on the model (Fig. S4), we computed the conditional R_c^2 and marginal coefficients of determination for generalized mixed-effect models (Nakagawa et al., 2017) with the R **MuMIn** package. The marginal one is interpreted as the variance explained by the deterministic effects R_{det}^2 . Secondly, R_c^2 expresses the variance explained by the full model, including stochastic and deterministic effects, thus $R_{stoc}^2 = R_c^2 - R_{det}^2$.

Empirical data

Data on species' populations are from two independent field studies conducted in different landscapes and replicated across multiple areas. Therefore, there is sufficient statistical power to incorporate the impact of time-varying environmental factors and disentangle deterministic and stochastic effects.

Annual plants dataset: The first study is located at Caracoles Ranch, an annual grassland of 2680 ha at Doñana National Park, southwest Spain (García-Callejas et al., 2021). The data gathered during > 360 hours of empirical observations associated with this study contains 9 years (2015 - 2023) of field surveys of local neighborhoods and species abundances following a spatial explicit design of 9 sites (8.5 m × 8.5 m along a 1 km × 200 m area) dominated by annual plant species with no perennial species present. In addition, each site was divided into 36 subplots of 1 m × 1 m with aisles of 0.5 m wide, for a total of 324 subplots and 20.412 data points. Soil conditions at these sites vary due to a salinity gradient and nutrient content. For our analysis, we considered the $n = 7$ plant species present every year, which belong to disparate taxonomic families and have different functional profiles (Table S1).

Wild bees dataset: The second study follows annual pollinator communities in shrublands situated in a landscape fragmentation gradient southwest of the Iberian Peninsula (Domínguez-García et al., 2024). Data on pollinator observations were collected during the flowering season over 8 consecutive years, with 7 – 10 rounds of data collection per year along a 100 m transect, totaling around 350 hours per year and 1.600 data points. To estimate pollinators' yearly abundance, we aggregated the number of times each pollinator species was recorded during the year. Note that we derived abundance from individual counts rather than visitation frequencies to avoid confounding pollinator activity with its frequency. We then normalized the abundances by the number of rounds—the sampling effort of that year—and considered the $n = 5$ species that encompass the 98% percentile of all counts across the 4 sites with more entries, which coincidentally have similar land use (Table S1).

Thanks to these unique datasets, we obtained empirical estimates of intrinsic growth rates $\tilde{r}_{t,b,i}$ that are variable in time t (across 9 and 8 years) plus an stochastic spatial effect across the different sampling sites b , and interactions $\tilde{A}_{t,ij}$ variable in time. Both datasets pertain to distinct evolutionary communities but are primarily subjected to the same environmental driver: annual rainfall. The plant community is driven by water availability, and the wild bees depend on flower production, which ultimately relies on water. Moreover, they have recently experienced drought periods (Fig. S2), making them suitable for studying the consequences of severe droughts on community structure. The anomalies for studying the effect of rainfall on stability and community composition are calculated as the difference from the

long-term 85-year average (520 mm).

Structural stability analysis

To connect community stability with rainfall-driven changes in species' intrinsic growth rates and in their interaction structure, we used the structuralist approach. It works on the principle that every possible community is feasible for a certain set of environmental conditions. Traditionally, environmental conditions are assumed to modify only species' performance (Rohr et al., 2014), and hence our work extends this approach by incorporating environmental-related variation in interactions too. Structural stability links the range of environmental conditions compatible with the persistence of multiple interacting species, the coexistence opportunities, with the structure of species interactions (Saavedra et al., 2017). The larger these coexistence opportunities, measured as the size of the so-called feasibility domain $\Omega(\tilde{A})$, the larger the range of species' intrinsic growth rates the community tolerates without losing species, Fig. S7. Originally, structural stability was defined only for competitive interactions and positive r (Saavedra et al., 2017). We normalized the (hyper)area of the convex hull defined by the column vectors of $\tilde{A}$, which determines the feasibility domain, by the area of the whole (hyper)sphere, rather than by the area defined by all the vectors with only positive elements. In this way, we could accommodate our situation, in which interactions are also positive and intrinsic growth rates can be negative.

The shape and size of the feasibility domain vary with the interaction network (Allen-Perkins et al., 2023; Grilli et al., 2017), which we show reorganizes with rainfall, Fig. 2. The position of $\tilde{r}$ also changes over time, as shown in (4). Then, for each rainfall condition, we lay in a different position within a feasibility domain that differs from that of the previous year. If the intrinsic growth rate vector is near a border of the feasibility domain, small shifts in it can drive some species to extinction (Allen-Perkins et al., 2023). So, as a proxy for stability, we measured how far $\tilde{r}$ lies from the borders for each year and each site (Fig. 4). We then averaged these distances across all sites. We adopted a sign convention in the distance to the borders: it is negative when $\tilde{r}$ is outside the feasibility domain. A graphical representation of these concepts is in Fig. S7.

There is a link between the structuralist approach and the Modern Coexistence Theory (Chesson, 2000; Godoy et al., 2018). The structural measure analogous to niche differences is precisely the size of the feasibility domain $\Omega(\tilde{A})$ since it quantifies coexistence opportunities (Saavedra et al., 2017). Similarly, the structural analog of fitness differences corresponds to the deviation between a given vector of intrinsic growth rates and the one that maximizes the possibility of a feasible solution, i.e., the incenter of the feasibility domain (Allen-Perkins et al., 2023). For quantifying these deviations in Fig. 3c, we measured the angle (in degrees) between the two vectors. We conducted these analyses, contributing to the R package anisoFun v2.0 [https:// github.com/RadicalCommEcol/anisoFun](https://github.com/RadicalCommEcol/anisoFun).

Network measures of temporal variability

We computed several properties of the interaction network structure and tracked how they varied with extreme environmental conditions. In particular, we tested the extent to which interactions were conserved over time, measuring the temporality of the networks between two times m and l (Bauzá Minguez et al., 2023):

$$\mathcal{T}_{m,l} = 1 - \frac{|\bigcap_{m,l}|}{|\bigcup_{m,l}|} \quad (5)$$

where $|\bigcup_{m,l}|$ is the size of the common links of the two networks, whereas $|\bigcap_{m,l}|$ is the size of the intersection (the links that appear only at one of the times). Then, temporality spans from 0 for networks with the same links, to 1 for completely different networks. It follows the same foundational concepts as beta diversity measures; while they describe the turnover of multiple species in space, $\mathcal{T}_{m,l}$ describes the

turnover of interactions in time. We first calculated the temporality of the unsigned networks, considering only interactions whose strength was within one standard deviation of the mean interaction strength of the corresponding year. Fig. S12 and S13 show those networks through time. The signed temporality deals with either positive or negative unweighted links, and its calculation and behavior are equivalent (Fig. S3).

To characterize the overall structure of the networks –i.e., including interactions’ signs and strengths–, we measured the percentage of species that meet:

$$|\tilde{A}_{t,ii}| \geq \sum_{j \neq i}^n |\tilde{A}_{t,ij}| \quad (6)$$

and tracked how it varied with environmental conditions. When the absolute intraspecific $\tilde{A}_{t,ii}$ is larger than the sum of the absolute interspecific interactions $\tilde{A}_{t,ij}$ for all species, the interaction matrix is diagonally dominant. This means that species limit themselves more than they limit other species. Note that classical studies may only compute the global difference between intraspecific and interspecific *competition*. This difference results in greater niche differentiation, and hence, in more opportunities for species to coexist (Chesson, 2000).

Simulations

Once we have fitted the parameters of (2), we can produce time series of abundances for a given rainfall value $\theta = (\theta_1, \dots, \theta_T)$, and then simulate how biodiversity unfolds under different rainfall regimes. The regimes in Fig. 5 were: constant rainfall equal to the 85-year average $\langle \theta \rangle = 520$ mm, oscillations between the minimum $\min(\theta) = 225$ mm and maximum $\max(\theta) = 625.5$ mm recorded rainfalls in our study period, increasing rainfall from $\langle \theta \rangle$ to $\max(\theta) + \sigma(\theta) \simeq 700$ mm, and decreasing rainfall and drought from $\langle \theta \rangle$ to $\min(\theta) - \sigma(\theta) \simeq 100$ mm. The simulations were run for the same temporal length as the annual plant and wild bees datasets, that is, $T = 9$ and $T = 8$ timesteps, respectively. We then calculate the Shannon index at each timestep t and site b as:

$$(\text{Shannon Index})_t^b = - \sum_i \left[\frac{N_t^{i,b}}{\sum_j N_t^{j,b}} \log \left(\frac{N_t^{i,b}}{\sum_j N_t^{j,b}} \right) \right] \quad (7)$$

Data and code availability

Datasets available at <https://doi.org/10.5281/zenodo.7527011> and <https://zenodo.org/records/10083504>. Historical rainfall data in the area (Aznalcázar station) was shared by [Meteoblue](#) (for the 85-year average) and [Junta de Andalucía](#). The code needed to reproduce this study can be downloaded from Zenodo <https://zenodo.org/records/15625585>.

Author contributions All authors conceptualized the ideas of the study. VC-S performed the data analysis. VC-S and OG drafted the manuscript with significant contributions from IB. OG provided the annual plant dataset, and IB provided the wild bee dataset.

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